

Mine Barnsongkit

New York, NY | sb5426@columbia.edu | supmine.github.io
linkedin.com/in/supanart-barnsongkit | github.com/supmine

EDUCATION

Columbia University

Expected December 2027

MS in Computer Science, Machine Learning track | New York, NY

Chulalongkorn University

August 2019 - May 2023

B.Eng. in Computer Engineering, **First Class Honours**, GPA: **3.73/4.00** | Bangkok, Thailand

Thesis: End-to-End Machine Learning Lifecycle Management Platform

Coursework: Computer Vision, Pattern Recognition, Data Structures and Algorithms

EXPERIENCE

Senior Software Engineer

May 2024 - August 2026

Arise by INFINITAS | Bangkok, Thailand

- Led five engineers building onboarding and account-opening services using Go, Kafka, Redis, MongoDB, and PostgreSQL.
- Owned deployment to Kubernetes with ArgoCD, go-live, monitoring with Prometheus and Grafana, and incident investigation.
- Reduced onboarding API response latency from approximately 2 seconds to 20 milliseconds through caching and MongoDB index optimization.

Software Engineer Team Lead

January 2023 - February 2024

Protomate | Bangkok, Thailand

- Led seven engineers across the smart helmet platform, Flutter app, and nRF-based remote control.
- Managed requirements, timelines, and roadmap alignment with mechanical and electrical engineering leads.
- Designed Bluetooth Classic and BLE connectivity, implemented Secure Boot, and integrated the app and remote with custom Android firmware.

Machine Learning Engineer Intern

June - July 2022

OxygenAI | Bangkok, Thailand

- Researched object detection in fisheye images without dewarping, addressing lens distortion and varying object orientations.
- Modified YOLOv5 with rotation-aware regression to correct bounding-box orientation.

PROJECTS

Rainfall Nowcasting on Thai DDS Radar Data

May 2022

Data Science and Engineering Project

- Built a cloud pipeline and dashboard with Google Cloud Functions, Pub/Sub, Cloud Storage, TensorFlow, Folium, and Power BI.
- Led optical flow, ConvLSTM, and RainNet modeling using TensorFlow and Keras; reported MSE below 0.001 on the project's radar and wind data.

Vehicle Detection and Counting

July 2021

Individual Research Study on Machine Learning

- Built a Mask R-CNN system to detect and segment vehicles and estimate traffic volume from CCTV data using PyTorch and OpenCV; reported 60.2% box AP50 and 56.6% mask AP50.

AWARDS & HONORS

RoboCup@Home - World First Runner-up

Open Platform League, 2022

- Helped build Walkie, a household service robot. Implemented ViLD-based pre-labeling for open-vocabulary object detection and assisted with person tracking and pose estimation. Eleven teams from seven countries qualified for the final.

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow, NumPy, pandas, Matplotlib, OpenCV

Languages: Python, Go, Kotlin, Java, C/C++, SQL (PostgreSQL), TypeScript, JavaScript

Frameworks: FastAPI, Gin, Fiber, Spring, React, Next.js, Node.js

Developer Tools: Git, Docker, AWS, Google Cloud Platform, Claude Code, Codex, VS Code